LONG-TERM EFFICACY AND SAFETY OF ADJUNCTIVE CENOBAMATE IN PATIENTS WITH UNCONTROLLED FOCAL SEIZURES: OPEN-LABEL EXTENSION OF A RANDOMIZED CLINICAL STUDY

Pavel Klein,[1] Gregory L. Krauss,[2] Sami Aboumatar,[3] Marc Kamin[4]

[1]Mid-Atlantic Epilepsy and Sleep Center, Bethesda, MD, USA; [2]Johns Hopkins University School of Medicine, Baltimore, MD, USA; [3]Austin Epilepsy Care Center, Austin, TX, USA; [4]SK Life Science, Inc., Paramus, NJ, USA

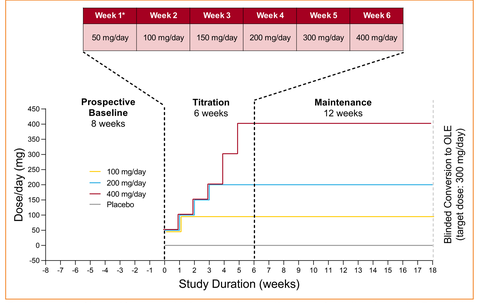
BACKGROUND

- A substantial portion of patients with epilepsy (>30%) do not achieve seizure freedom despite availability of newer antiepileptic drug (AED) therapies[1,2]
- Clinical studies have revealed substantial reductions in seizure frequency and high responder rates, including 100% seizure reduction, in cenobamate-treated adult patients with uncontrolled focal seizures[3,4]
- In the 18-week (6-week titration/12-week maintenance) double-blind (DB), dose-response phase 2 study (YKP3089C017, C017), the median percent reduction in focal seizure frequency per 28 days was 35.5%, 55.0%, and 55.0% for cenobamate (CNB) 100, 200, and 400 mg, compared with 24.0% for placebo (PBO; all P<0.01 vs PBO)[3]
  - Notably, 3.9%, 11.2%, and 21.1% of CNB 100, 200, and 400 mg patients, respectively, were seizure-free during the 12-week maintenance treatment phase, compared with 1.0% for placebo[3]
- The most common treatment-emergent adverse events (TEAEs) with CNB during C017 DB treatment were central nervous system–related (eg, somnolence, dizziness)
- Patients who completed the entire 18-week DB treatment period were eligible to continue CNB in an open-label extension (OLE)
- Here we report long-term efficacy and safety outcomes from the ongoing C017 OLE

DESIGN/METHODS

- Patients in the multicenter, multinational, DB, randomized, PBO-controlled C017 study (ClinicalTrials.gov NCT01866111) were randomized 1:1:1:1 to PBO or CBN 100, 200, or 400 mg once daily (**Figure 1**)
- Key inclusion criteria:
  - Adults 18-70 years old with uncontrolled focal epilepsy
  - Taking 1-3 AEDs at stable doses for at least 4 weeks before screening
  - Patients must have experienced ≥8 seizures during baseline and had to have ≥3 focal seizures during each of the 2 consecutive 4-week baseline periods and no consecutive 25-day seizure-free interval
- Key exclusion criteria included history of:
  - Diazepam, phenytoin, or phenobarbital use within 1 month of screening due to potential for drug-drug interaction
  - Receiving vigabatrin within the past year, felbamate for less than 18 continuous months, or intermittent rescue benzodiazepines more than once a month within the past year
  - Status epilepticus
  - Clinically significant psychiatric illness or suicidal ideation
  - Drug or alcohol abuse
  - Serious hypersensitivity drug reaction

Figure 1. C017 Study Design



*The initial starting dose of the original, faster titration schedule was 100 mg/day with weekly increments of 100 mg/day to the target dose. The amended titration schedule reduced the initial starting dose to 50 mg/day and slowed the titration rate to the target dose to improve tolerability.
OLE, open-label extension.

- A total of 360 patients completed the DB study and were eligible to enter the OLE
- Patients entering the OLE underwent a 2-week blinded conversion to a target dose of CNB 300 mg once daily
  - During the 2-week conversion, the investigator could increase or decrease the open-label dosage, if clinically indicated, to a minimum of 50 mg and maximum of 400 mg/day
- Doses of concomitant AEDs could be adjusted during the conversion phase
- During the OLE treatment phase, concomitant AEDs could be added, removed, or adjusted (no cenobamate monotherapy allowed)
- Scheduled study assessments occurred every 3 months
- The study is ongoing

ANALYSIS

- This analysis used a data cut-off of April 23, 2018
- The OLE safety population was defined as all patients who entered the OLE and had taken at least 1 dose of CNB
- The OLE intent-to-treat (ITT) population was defined as all patients who had taken at least 1 dose of CNB and had any seizure data recorded in the OLE

RESULTS

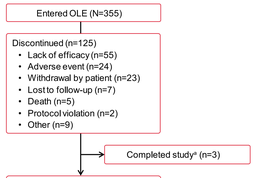
Patients

- 98.6% (355/360) of patients who completed the DB study entered the OLE (OLE safety population, N=355; OLE ITT population, n=354)
- Of these 355 patients, 265 were originally randomized to CNB (CNB/CNB) and 90 were originally randomized to PBO and transitioned to CNB (PBO/CNB)
- Baseline demographics of the OLE were similar to those of the DB study with regards to median age (38 years, range: 19-70), sex (47.9% female), and mean body weight (77.1 kg)
- The majority of patients in the OLE population (84.5% [300/355]) were taking 2 or 3 concomitant AEDs at baseline of the DB study

Exposure

- As of April 2018, 63.9% (227/355) of patients were continuing in the OLE; 125 patients discontinued, including 15.5% (55/355) due to lack of efficacy, 6.8% (24/355) due to adverse events, and 6.5% (23/355) due to withdrawal by patient (**Figure 2**)

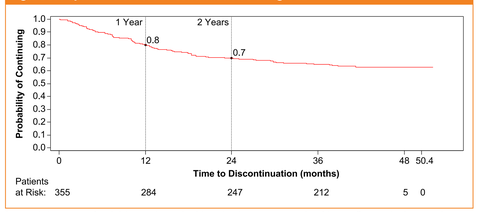
Figure 2. Patient Disposition and Reason for Discontinuation (OLE Population)



[a]As reported on the end of study case report form. OLE, open-label extension.

- The median duration of CNB OLE exposure was 40.1 months (range: 1.1-54.4 months)
- 1 year after OLE initiation, 80% of patients continued CNB treatment; at 2 years, 70% remained (**Figure 3**)

Figure 3. Kaplan-Meier Plot of Continuation During OLE

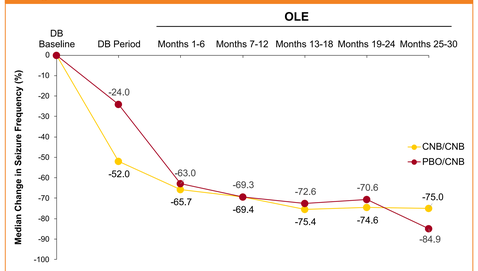


Event = early discontinuation from OLE. Patients who completed the study and patients ongoing at date of data cut-off are considered as censored.
OLE, open-label extension.

Efficacy

- The median modal daily CNB dose was 300 mg (range: 50-400 mg)
- Median percent reduction in seizure frequency during the first 6 months of the OLE for all CNB OLE patients was 65.4% and was similar among patients originally treated with CNB or PBO in the DB study (**Figure 4**)
- The median percent reduction in seizure frequency over baseline for all CNB OLE patients increased with each 6-month OLE interval, up to 76.0% at 25-30 months

Figure 4. Median Percent Change in Seizure Frequency/28 Days by 6-Month Intervals During OLE (ITT Population, n=354)

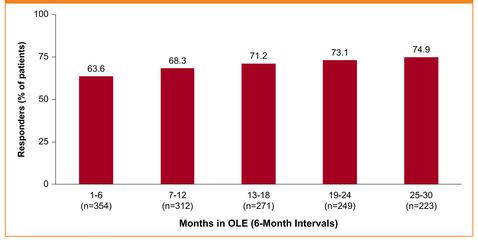


Group, n	DB Phase	Months 1-6	Months 7-12	Months 13-18	Months 19-24	Months 25-30
CNB/CNB	328	264	232	201	185	163
PBO/CNB	106	90	80	70	64	60

Median baseline seizure frequency/28 days during the DB study for the 354 patients in the ITT population was 9.5 (range, 3.5-704.0).
CNB, cenobamate; DB, double-blind, ITT, intent-to-treat, OLE, open-label extension; PBO, placebo.

- ≥50% responder rates (percentage of patients with ≥50% seizure reduction) also increased with each 6-month OLE interval, up to a median 74.9% for all CNB OLE patients at 25-30 months (**Figure 5**)

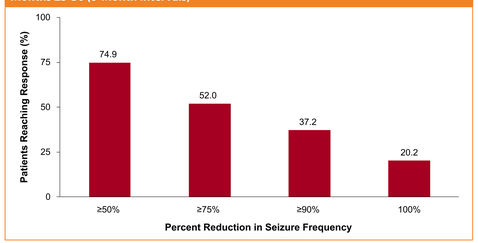
Figure 5. ≥50% Reduction in Seizure Frequency by 6-Month Intervals During OLE (ITT Population, n=354)



ITT, intent-to-treat; OLE, open-label extension.

- At Months 25-30, 20.2% (45/223) of evaluable patients were seizure-free (**Figure 6**)
- Among the 45 patients who were seizure-free (100% seizure reduction) at Months 25-30, the median duration of seizure freedom achieved during the entire OLE was 33.2 months (range, 13.2-50.4 months)
- When analyzed at Months 28-30 (3-month intervals), seizure frequency reductions of ≥50%, ≥75%, ≥90%, and 100% were achieved in 76.5%, 55.4%, 36.2%, and 25.4% of patients, respectively
  - The median modal dose among seizure-free patients during this time was 300 mg

Figure 6. ≥50%, ≥75%, ≥90%, and 100% Reduction in Seizure Frequency at Months 25-30 (6-Month Intervals)



Safety

Deaths and Serious TEAEs

- At data cut-off, five deaths had been reported in the OLE (pneumonia/sepsis, septicemia, fatal injuries after being struck by a car, cardiogenic shock, and myocardial infarction)
  - All were considered to be unrelated to study drug
- Serious TEAEs occurred in 18.3% (65/355) of patients; seizure (1.4%, n=5) and vertigo (1.1%, n=4) were the only serious TEAEs reported in >1% of patients (**Table 2**)

Treatment-Emergent Adverse Events (TEAEs)

- TEAEs occurred in 87.6% (311/355) of patients during the OLE (**Table 2**)
  - TEAES reported in ≥10% of patients in any group were dizziness, somnolence, fatigue, diplopia, headache, gait disturbances, and vertigo

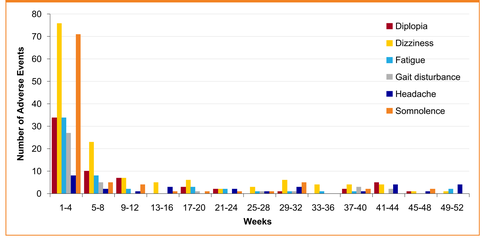
Table 2. Summary of TEAEs (Safety Population)

n, %	All CNB (n=355)	CNB/CNB (n=265)	PBO/CNB (n=90)
Patients with ≥1 TEAE	311 (87.6)	233 (87.9)	78 (86.7)
Patients with ≥1 serious TEAE	65 (18.3)	49 (18.5)	16 (17.8)
Patients with ≥1 TEAE leading to discontinuation	28 (7.9)	19 (7.2)	9 (10.0)
TEAEs in ≥10% in any group			
Dizziness	116 (32.7)	87 (32.8)	29 (32.2)
Somnolence	85 (23.9)	53 (20.0)	32 (35.6)
Fatigue	53 (14.9)	39 (14.7)	14 (15.6)
Diplopia	49 (13.8)	35 (13.2)	14 (15.6)
Headache	42 (11.8)	33 (12.5)	9 (10.0)
Gait disturbances	39 (11.0)	30 (11.3)	9 (10.0)
Vertigo	30 (8.5)	17 (6.4)	13 (14.4)

CNB, cenobamate; PBO, placebo; TEAEs, treatment-emergent adverse events.

- TEAEs leading to discontinuation occurred in 7.9% (28/355) of patients, most frequently due to nervous system disorders (3.1%, n=11), with dizziness (0.8%, n=3) and somnolence (0.6%, n=2) the most frequently (≥0.5%) reported
  - Other categories of TEAEs leading to discontinuation that included more than 1 patient were skin and subcutaneous tissue disorders (1.7%, n=6); psychiatric disorders (1.1%, n=4); infections (0.6%, n=2); and eye disorders (0.6%, n=2)
- In the OLE, TEAEs most frequently occurred during the first month of treatment (**Figure 7**)

Figure 7. Time of Onset of the Most Common TEAEs During OLE Treatment



OLE, open-label extension; TEAEs, treatment-emergent adverse events.

CONCLUSIONS

- CNB was generally well tolerated during long-term treatment (up to 54 months), with 64% of patients continuing treatment as of April 2018, and 70% of patients continuing for at least 2 years
- Reductions in seizure frequency were sustained during 30 months of adjunctive CNB treatment
- 20.2% of evaluable patients were seizure-free at 25-30 months

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